

Mohiuddin Bilwal

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[LinkedIn](#) | [Portfolio](#) | [GitHub](#) | [Medium](#) | [ResearchGate](#) | [Google Scholar](#)

RESEARCH INTERESTS

Machine and Deep Learning, Large Language Models, Model Compression Techniques, Natural Language Processing, Human-Computer Interaction, Software Engineering, Code Vulnerabilities, Query Optimization, System Architecture and Software Security

EDUCATION

BSc in Computer Science and Engineering

Jan 2020 – Apr 2024

Military Institute of Science and Technology, Dhaka, Bangladesh

Thesis: "Simulating User Behavior to Evaluate the Navigability of Websites Through Artificial Intelligence"

CGPA: 3.41 / 4.00

TEST SCORE

IELTS Academic: 7.0 (Listening: 7.0, Reading: 7.0, Writing: 6.5, Speaking: 6.5)

Oct 2025

RESEARCH EXPERIENCE

Researcher

Aug 2025 – Jul 2026

SwAPP Lab, Iowa State University | Supervisor: Prof. Ali Jannesari

- Performed structured pruning on CNN and ViT architectures of ProfilingAgent research, cutting memory by up to 7.7% while keeping accuracy loss below 1%.
- Reproduced structured pruning and quantization on CNN and ViT architectures (GETA), achieving up to 61% parameter and 44% MACs reduction while matching published baselines within 0.02%.
- Reviewed literature, built result tables, and fine-tuned models for an LLM-oriented model compression study.

PUBLICATIONS

- [1] S. Jafari, A. Sarkar, **M. Bilwal**, A. Jannesari, "ProfilingAgent: Profiling-Guided Agentic Reasoning for Adaptive Model Optimization," *IEEE Computational Intelligence Society (Under Review)*. [\[preprint\]](#)
- [2] **M. Bilwal**, S. Abrar, Z. Sultana, J. U. Tanvin, R. M. R. Shahriar, M. M. Rahman, M. N. Islam, "Proposing and Evaluating a Modified Keyboard Layout for One-Handed Users." *27th International Conference on Computer and Information Technology (ICIT 2024), Cox's Bazar, Bangladesh*. [\[ref\]](#)
- [3] **M. Bilwal**, J. U. Tanvin, S. M. S. Rahman, M. R. F. Faisal, A. F. Khan, S. E. Arafat, M. Akhtaruzzaman, "A Multi-Modal Assistive Robot Navigation System for Physically Impaired Users." *IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2025), Bangladesh Army University of Science and Technology (BAUST), Saidpur, Bangladesh*. [\[ref\]](#)
- [4] A. Hossain, **M. Bilwal**, A. Sircar, M. N. Islam, "Simulating User Behavior to Evaluate the Navigability of Websites Through Artificial Intelligence." *In Preparation, 2026*. [\[ref\]](#)
- [5] F. N. Nizami, **M. Bilwal**, A. Sircar, B. Ghosh, M. Akhtaruzzaman, S. R. Khan, N. N. Khanom, "Detection of Corona Virus Using CBC Blood Information." *Draft*. [\[ref\]](#)

TECHNICAL SKILLS

Programming: Python, Java, Ruby, Golang, C++, ReactJS, Rails, SQL, Redis, REST API, FAST API

ML / NLP: PyTorch, TensorFlow, Hugging Face, spaCy, NLTK, RAG, LLM, scikit-learn, CNN, Vision Transformers

Tools: Git, GitHub, Docker, Gateway, POSTMAN, LaTeX, Linux, Microsoft Azure, AWS S3 Bucket

Embedded & IoT: Raspberry Pi, Arduino, GPU

Other Knowledge: Background Jobs, Gateways, Scheduler, Slurm, Pruning, Quantization, Distillation, OpenRouter API

PROJECTS

Agentic Profiling-Guided Pruning and Quantization Framework (Ongoing Research)

Tech Tools: Python, PyTorch, Pruning, Quantization-Aware Training (QAT), Knowledge Distillation, OpenAI API, CUDA

- Developed an agent based framework for model compression using LLM-guided structured pruning, adaptive fine-tuning, and mixed-precision quantization-aware training.
- Designed the ProfilingAgent to analyze model architectures, estimate pruning sensitivity, and generate profiling information that guides compression decisions.
- Evaluated the framework on ResNet, ViT, DeiT, and Swin models using ImageNet and CIFAR-10, targeting high compression with minimal accuracy degradation.

NextGen Wheelchair: A Semi-Autonomous Solution for Disabled People Using Head Motion

[\[Link\]](#)

Tech Tools: OpenCV, Machine Learning, Bootstrap, JavaScript, JSON

- Developed and manufactured a wheelchair prototype for individuals with mobility issues to drive independently.
- Integrated head motion, voice, joystick, and sonar-sensor-based control and alert systems as key features.
- Implemented a web platform for caregivers to manually operate the prototype via the web interface.

CVCBC Predictor: Detection of Coronavirus Using CBC Blood Information of Patients

[\[Link\]](#)

Tech Tools: Machine Learning

- Investigated the advantages of using Complete Blood Count (CBC) alongside PCR in COVID-19 diagnostics by analyzing CBC data of 4474 patients in Brazil.

Air Quality Prediction of Nairobi

Tech Tools: Machine Learning, AutoRegression Model, MongoDB

- Worked on Africa's largest open data platform, "openAfrica" in WorldQuant University's virtual Data Science Lab.
- Prepared time series data and built an autoregression model to predict PM 2.5 readings throughout the day.
- Tuned hyperparameters to improve model performance.

ACHIEVEMENTS

1. Completed the AI-Integrated Leadership Program (AILP), Cohort 01, 2026, offered by Aspire Institute. [\[Link\]](#)
2. Completed the prestigious Aspire Leaders Program 2025 (Cohort 02), funded by Aspire Institute, founded at Harvard University. [\[Link\]](#)
3. Achieved 3rd position in the internal Hackathon Competition at Barikoi Maps, 2025.
4. Participated in the National Collegiate Programming Contest (NCPC), Jahangirnagar University, 2024.
5. Achieved 24th position out of 70 teams in the ML Olympiad - CO2 Emissions Prediction Challenge, Kaggle, 2024.
6. Participated in the International Collegiate Programming Contest (ICPC), Asia Dhaka Regional, BUBT, 2023.
7. Achieved 34th position out of 100+ teams in LEETCON (Cybersecurity Competition), MIST, 2023.
8. Participated in the Inter University Programming Contest (IUPC), SUST, 2023.
9. Two abstracts [\[Link\]](#) got accepted in the IEEE CS BDC Summer Symposium, Islamic University, Kushtia, 2023.
10. Achieved 1st position out of 36 universities worldwide in the University Rover Challenge (URC) 2021.

TEACHING EXPERIENCE

Mentor, Team Coordinator — MIST Computer Club, MIST Innovation Club

Jan 2021 - Mar 2023

Military Institute of Science and Technology (MIST)

- Conducted sessions on different algorithms and competitive programming problems.
- Mentored junior programmers and assisted in problem-solving sessions.
- Conducted workshops on basic web development languages to give newly admitted students an overview of web development.

PROFESSIONAL EXPERIENCE

Junior Software Engineer — Research & Development Department

Jul 2025 - Present

TechnoNext Software Limited, Dhaka, Bangladesh

- Designed and optimized high-traffic REST APIs, reducing response times by up to 60–70% through query optimization, caching strategies, and efficient data retrieval techniques.
- Resolved critical N+1 query issues and optimized complex ActiveRecord queries using eager loading, indexing, and database performance tuning techniques.
- Built and enhanced multi-tenant backend architecture with modular service-oriented design, strengthening authorization and authentication management, maintainability, and long-term scalability.
- Implemented comprehensive automated testing using RSpec across controllers, models, services, background jobs, and request specs to improve system reliability and deployment confidence.
- Investigated and resolved complex production issues involving authentication, rate limits, session management, caching layers, API integrations, and legacy backend systems.
- Contributed to subscription billing and payment gateway integration research and implementation, helping design reliable and extensible payment workflows for SaaS applications.

Tech Stack: Ruby on Rails, PostgreSQL, Redis, React.js, REST API, Docker, Gateway

Software Engineer Intern

May 2025 - Jul 2025

Barikoi Maps, Dhaka, Bangladesh

- Solved algorithmic challenges and applied competitive programming skills to optimize geospatial mapping solutions.
- Built tools, scripts, and internal APIs to automate and support development workflows.
- Worked with data structures, parsing systems, and backend logic.
- Collaborated with senior engineers and participated in code reviews.
- Learned best practices in software development, testing, and deployment.

Tech Stack: Python, PostgreSQL, FAST API, JavaScript, Networking

CERTIFICATIONS

1. Udacity: Introducing Generative AI with AWS. [\[Link\]](#)
2. Microsoft Certified: Azure AI Fundamentals [\[Link\]](#)
3. LinkedIn: Project Management Foundations [\[Link\]](#)

VOLUNTEER EXPERIENCE

1. Volunteered in the Event Management Committee at the National Collegiate Programming Contest (NCPC) 2020, organizing logistics including decorations, food distribution, and information dissemination for participating teams.
2. Organized events in the Campus Hour program, including Treasure Hunt, on behalf of MIST Computer Club (MCC).
3. Worked with the Event Management Committee to organize the Intra-Department Short Pitch Cricket and Football Tournament, 2024.
4. Worked as an Open Source Contributor in GirlScript Summer of Code, 2023.

REFERENCES

1. Prof. Lt. Col. Muhammad Nazrul Islam
Associate Professor
Computer Science and Engineering (CSE)
Military Institute of Science and Technology (MIST)
Email: nazrul@cse.mist.ac.bd

2. Prof. Md. Akhtaruzzaman
Associate Professor and Head
Robotics and Mechatronics Engineering
Daffodil International University (DIU)
Email: akhtaruzzaman.cse@diu.edu.bd